E-WASTE CLASSIFICATION & RECYCLING SYSTEM

Quick Setup Guide & System Prerequisites for Peers and Evaluators

1. System Prerequisites (Must Be Installed)

Python 3.10 - 3.13

Required for the Flask AI backend.
IMPORTANT: Check 'Add Python to PATH' during installation.
Download: https://www.python.org/downloads/

Node.js (LTS Version)

Required for the Vite React frontend.
Includes npm automatically.
Download: https://nodejs.org/

VS Code

Recommended code editor.
Install the 'Python' extension (by Microsoft)
from the Extensions tab (Ctrl+Shift+X).

2. How to Run the Project in VS Code (Step-by-Step)

Step A: Extract & Open Folder

1. Extract 'e_waste_classification_project.zip'.
2. In VS Code, go to File > Open Folder... and select the extracted folder.

Step B: Launch Backend (Terminal 1)

Open terminal (Ctrl + `) and run:

```
    cd backend
    pip install -r requirements.txt
    python app.py
=> Output: Running on http://localhost:5000 (or double-click run_backend.bat)
```

Step C: Launch Frontend (Terminal 2)

In VS Code, click '+' in terminal to open a 2nd terminal tab, then run:

```
    cd frontend_upload/frontend
    npm install
    npm run dev
=> Output: Local: http://localhost:5173/
```

Step D: Open in Browser

Go to http://localhost:5173 in Chrome, Edge, or Firefox.
Upload an image of an electronic item and click 'Analyze image'.

SUPPORTED CATEGORIES & SYSTEM ARCHITECTURE

Model Capabilities, Disposal Protocols, and Troubleshooting

3. Supported 6 E-Waste Categories

Mobile Phone Scope: Smartphones, iPhones, Androids, cracked screens	**Data wipe, recovery of precious metals (Au, Ag, Cu)**
Laptop Scope: Notebooks, MacBooks, ThinkPads, keyboards	**Component salvage (RAM, SSD), IT asset certified recycling**
Battery Scope: Li-ion cells, AA/AAA, 9V, power banks	**FIRE HAZARD: Never trash; tape terminals; hazardous waste depot**
Charger / Cable Scope: Power adapters, USB cables, charging bricks	**Reuse with adapter converters; recovery of copper wiring**
Circuit Board Scope: PCBs, motherboards, GPUs, electronic scrap	**Smelter recovery for rare earths, gold, and palladium**
Monitor / TV Scope: LCD/LED displays, televisions, CRT screens	**Contains lead/mercury; municipal collection center drop-off**

4. Technical Architecture (For Evaluation / VIVA)

- Model: Dual-Backbone Transfer Learning Ensemble (EfficientNetB0 + MobileNetV2)
- Vision Layer: Multi-scale semantic synset mapping + aspect-ratio geometry heuristics
- Substrate Verification: FR-4 green/blue color substrate and edge density analyzer
- Accuracy: Calibrated 91% - 98% confidence distribution with top-3 alternative ranks
- Backend: Python Flask REST API (/predict, /health, /classes) on port 5000
- Frontend: React 18 + Vite with dynamic category override chips on port 5173

5. Common Troubleshooting

- Port 5000 in use: In PowerShell run: Get-Process python | Stop-Process -Force
- 'python' command not found: Re-run Python installer and check 'Add to PATH'.
- 'npm' command not found: Ensure Node.js is installed from nodejs.org and restart VS Code.